

Mend-Amar Badral

 MendeBadra |  Mend-Amar Badral |  mendebadra.github.io
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Driven by a problem solving mindset and fueled by curiosity, I always aspire to develop solutions for the real-world, impactful problems.

EDUCATION

Budapest University of Technology and Economics (BME) Sep, 2025 – June, 2027
M.Sc. in Computer Science Engineering Budapest, Hungary
Supervisor: [Márton Vaitkus](#)
Specialization: Data Science & Artificial Intelligence

National University of Mongolia (NUM) Sep, 2021 – June, 2025
B.Sc. in Applied Mathematics Ulaanbaatar, Mongolia
GPA: 3.2/4.0 (**rank 3/34**)
Supervisor: [Galtbayar Artbazar](#)
Thesis: *Evaluating land degradation using image processing methods*

EXPERIENCE

Center of Mathematics Application, NUM Oct, 2023 – June, 2025
Undergraduate Research Intern

- **Lab Lead:** Selected as Undergraduate Lab Lead amongst the 6-8 student team in the final year. Co-taught and participated in weekly seminar meetings conducted by the research center.
- **Tool development:** Developed a custom, vendor specific **software** which takes in drone camera images and sensor parameters to calculate a vegetation index. Implemented a command line interface for the tool such that it can process images in custom folder structure in Linux environment. This change significantly reduced the time spent on processing drone imagery.
- **Organization:** Implemented a unified, documented folder structure for the research data backup which enabled efficient data retrieval for research purposes and easy integration of data to be collected in the future.
- **Contributions:** Incorporated an open-source Docker containerized application (OpenDroneMap) for drone orthomosaic processing instead of proprietary software.

PROJECTS

[aiSim Data Integration for 3D Gaussian Splatting](#)

- Implemented a JSON parser to convert sensor measurements from the [aiSim](#), a simulation software, into a 3D scene representation software ([nerfstudio](#)) format.

[Deep Learning Foundations and Concepts Book Examples Reproduction](#)

- Reproduced examples and figures from C. Bishop's *Deep Learning Foundations and Concepts* book.

SKILLS

Programming: Python, Bash, Julia, C/C++, R, MATLAB
Tools: Docker, Linux, Git, GitHub, GitHub Actions, L^AT_EX